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Metafluidic shaping: Simultaneous flow uniformity and drag reduction

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Abstract – Disordered flow patterns and elevated resistance at abrupt geometric transitions are critical issues in systems from blood vessels to microfluidics, where they are linked to disease and inefficiency. Conventional flow control strategies face a fundamental dilemma: efforts to reduce drag often disrupt flow uniformity, while attempts to rapidly stabilize the flow typically increase resistance. Here, we introduce a metafluidic shaping strategy that overcomes this limitation, enabling simultaneous drag reduction and rapid flow stabilization. Inspired by metamaterial concepts, we treat the geometric transition as a dipole scatterer and deploy auxiliary structures to generate a counteracting flow field, effectively canceling adverse hydraulic effects. Experimental visualizations confirm the complete suppression of local recirculation, leading to highly uniform flows with substantially reduced drag. Furthermore, applying this technique to liquid cooling plates significantly enhances thermal performance, yielding lower and more homogeneous temperatures. Our work demonstrates that a metamaterial approach can overcome fundamental limitations of traditional fluid dynamics.

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Introduction. – Disordered blood flow and increased vascular resistance significantly contribute to cardiovascular disease (fig. 1(a)) [1–3]. According to the Centers for Disease Control and Prevention (CDC), heart disease is the leading cause of death in the U.S. Efforts to reduce flow resistance and create uniform flow often conflict. For instance, increasing vessel diameter or decreasing blood viscosity lowers flow resistance but raising the local Reynolds number at sites of vascular cross-sectional variation. This increases the risk of flow separation, turbulence, and vortex formation, disrupting stable laminar flow and potentially leading to pathological changes.

In laminar pipe flow, resistance primarily includes frictional losses and local head loss at expansions, contractions, or junctions [4–7]. Velocity redistribution at interfaces of varying channel sizes leads to viscous energy dissipation, causing significant energy loss. Current strategies to minimize local head loss, often based on empirical designs like gradual transitions or small-angle diffusers, only partially address flow smoothing and separation prevention [8]. Completely eliminating these losses is theoretically impossible. Achieving a rapid and uniform flow field is another major challenge. Methods like adding branches or optimizing fractal structures improve flow uniformity but typically increase resistance [9–11]. Both theory and practice suggest that reducing flow resistance and achieving uniform flow are inherently conflicting objectives.

Recent advances in the field of hydrodynamic metamaterials have brought new possibilities for addressing these challenges [12–21]. Through the deliberate design of artificial structures, it is now possible to precisely manipulate fluid flow, enabling previously unattainable functionalities [22–28]. In the realm of drag reduction,

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numerous studies have explored the use of cloaking strategies to minimize flow resistance [29,30]. In the context of achieving rapid flow uniformity, Chen *et al.* introduced the concept of a "hydrodynamic evener", which leverages the theory of scattering cancellation in porous media to promote rapid flow homogenization during channel transitions [31]. However, such porous-media–based designs are associated with certain inherent challenges, including increased flow resistance, complex manufacturing requirements, and stringent fluid cleanliness standards. In applications such as blood vessels or liquid cooling plates, the design objective is to optimize fluid behavior within laminar flow channels, rather than within dense porous matrices. Consequently, fundamentally different strategies are required for these scenarios.

Here, we introduce a "metafluidic shaping" strategy —a method of manipulating fluid flow rooted in hydrodynamic metamaterials— for shaping flow at abrupt cross-sectional transitions. Recognizing that sudden pipeline expansions act as "obstacles" that scatter incoming flow, we approximate the primary scattering contribution as a dipole. Physically, this dipole arises from the high-pressure stagnation region upstream of the expansion (a source-like effect) and the low-pressure recirculation zone downstream (a sink-like effect). We then strategically introduce auxiliary structures that generate reverse scattering to actively cancel these adverse effects [32], as shown in fig. 1(b). Unlike conventional passive minimization approaches or porous media-based designs, our method directly engineers channel geometry in critical regions to manipulate velocity profiles in the main flow passage, as shown in fig. 1(c). Numerical simulations demonstrate remarkable zero local pressure drop performance: beyond intrinsic channel resistance, additional pressure losses typically induced by geometric transitions are entirely eliminated (fig. 1(d)). Experimental validation confirms that the metadevice can rapidly and uniformly distribute flow while reducing flow resistance and effectively suppressing expansion-induced recirculation regions —an essential feature for blood flow management. Existing stents often not only fail to manage flow at vessel junctions but can even promote recirculation. Additionally, experiments demonstrate that applying this strategy to aluminum alloy cooling plates achieves excellent thermal management by both reducing temperature and enhancing temperature uniformity. Therefore, the presented approach serves as a versatile platform technology, offering immediate benefits for applications like microelectronics cooling [33,34] and battery thermal management [35–37], while also providing a flexible framework for integration into complex multiphysics [38–40] problems.

Governing equation of fluid dynamics. – We consider fluid flow in a narrow gap between two parallel plates, where the gap height $h(x, y)$ varies slowly with the in-plane coordinates (x, y). Under the lubrication approximation ($h \ll L$, where L is a characteristic in-plane length scale,

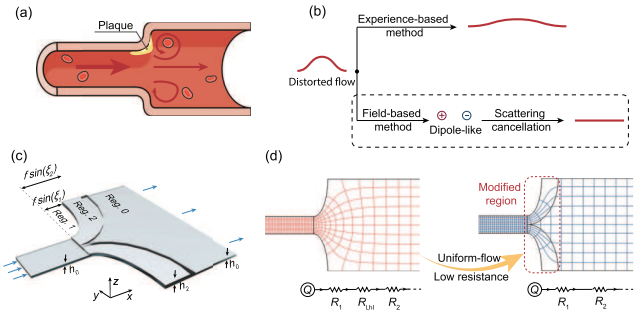

Fig. 1: Concept and prospects of the metadevice designed via scattering cancellation theory. (a) Variations in vessel cross-sectional area inevitably induce non-uniform flow fields, leading to phenomena such as flow recirculation and increased hydraulic resistance. (b) Experience-based design methods can mitigate but not completely eliminate distored flow. Field-based approaches hold promise for addressing this challenge. (c) Schematic illustration of the metadevice proposed in this study. (d) Comparison of flow fields between a normal device and the metadevice. The metadevice rapidly establishes a uniform flow field and effectively eliminates local head loss.

and $|\nabla h| \ll 1$), the flow is predominantly in the x-y plane (Hele-Shaw model). The depth-averaged velocity $\bar{\mathbf{u}}(x, y)$ is related to the pressure gradient $\nabla_{x,y} p$ by Darcy's law for flow in a Hele-Shaw cell:

$$\bar{\mathbf{u}}(x, y) = -\frac{h^2(x, y)}{12\mu} \nabla_{x,y} p(x, y), \tag{1}$$

where μ is the dynamic viscosity. The volumetric flux per unit width, $\mathbf{q}(x, y) = \bar{\mathbf{u}} h$, is then

$$\mathbf{q}(x, y) = -\frac{h^3(x, y)}{12\mu} \nabla_{x,y} p(x, y). \tag{2}$$

For an incompressible fluid, the continuity equation $\nabla_{x,y} \cdot \mathbf{q} = 0$ leads to a Laplace-like equation for pressure:

$$\nabla_{x,y} \cdot (D(x, y) \nabla_{x,y} p(x, y)) = 0, \tag{3}$$

where $D(x, y) \equiv h^3(x, y)/(12\mu)$ is the effective hydraulic conductivity or "diffusivity."

This equation forms the basis for applying scattering cancellation theory (SCT), as it is analogous to the governing equations in electrostatics or steady-state heat conduction where SCT is well-established.

Design principles and process of the metadevice. – To achieve a uniform flow in the central rectangular chamber, the expander sections are designed as quarter-elliptical shells, as shown in fig. 1(c). We aim to cancel the lowest-order scattering mode (dipole scattering) generated by these shells when exposed to an incoming uniform flow. This focus on the dipole mode is deliberate: elliptical (or circular) profiles primarily produce dipole scattering under the Hele-Shaw approximation, simplifying the cancellation process, whereas arbitrary shapes may generate higher-order modes (*e.g.*, quadrupoles) that complicate flow control. The structure consists of three regions in elliptical

coordinates (ξ, η): an inner core region $(0 \leq \xi < \xi_1)$ assumed to have zero flow (*e.g.*, $p = 0$ or solid boundary), the elliptical shell $(\xi_1 < \xi < \xi_2)$ with diffusivity D_2, and the outer background region $(\xi > \xi_2)$ with diffusivity D_0 subjected to a far-field pressure gradient $G = -\nabla p_x^\infty$.

The detailed derivation (see the Supplementary Material `Supplementarymaterial.pdf` (SM)[1], sect. S1), involving solving eq. (3) in elliptical coordinates for each region and applying appropriate boundary conditions for pressure and normal flux continuity, shows that the dipole scattering coefficient can be made zero if the diffusivity of the shell D_2 satisfies

$$D_2 = D_0 \coth \xi_2 \coth(\xi_2 - \xi_1). \tag{4}$$

Since $D \propto h^3$, this condition dictates a specific ratio for the channel heights:

$$\frac{h_2}{h_0} = (\coth \xi_2 \coth(\xi_2 - \xi_1))^{1/3} > 1, \tag{5}$$

where h_2 and h_0 are the channel heights in the shell and background regions, respectively. This implies that the shell region must have a increased height compared to the background to achieve flow homogenization.

Numerical simulation of the metadevice flow field. – To verify that the designed expansion-contraction metadevice possesses the characteristics of zero local head loss and uniform flow, we first conducted simulations using COMSOL Multiphysics. The selected geometrical parameters of the device were $\xi_1 = 0.4$, $\xi_2 = 0.8$, $f = 3\,\text{mm}$ and $h_0 = 1\,\text{mm}$, where f is the semi-focal distance. According to eq. (4), h_2 should be 1.58 mm; however, considering the flow losses in the expansion channel, the optimized value of h_2 was set to 1.59 mm. The overall length and width of the device were $L = 13\,\text{cm}$ and $W = 8\,\text{cm}$, respectively. The inlet boundary condition was imposed as a flow rate of 1 ml/s, while the outlet was maintained at a pressure of zero. All other boundaries were set as no-slip conditions.

Figures 2(a)–(c) present the simulated results of streamlines, velocity, and pressure distributions at $z = 0.5\,\text{mm}$, while figs. 2(d)–(f) illustrate the corresponding results for the normal device. The results clearly demonstrate that the metadevice achieves highly uniform flow, in stark contrast to the "normal device" with a simple sudden expansion. In the metadevice, streamlines are smoothly guided from the narrow inlet to the wider central chamber, forming nearly parallel trajectories. The velocity profile across the entire chamber width is remarkably uniform, closely approximating the characteristics of one-dimensional Poiseuille flow. The pressure contours are smooth and evenly spaced, indicating a gradual pressure drop without adverse gradients that could induce flow separation. The above conclusions demonstrate strong robustness across various scenarios. In sect. S2 of the SM,

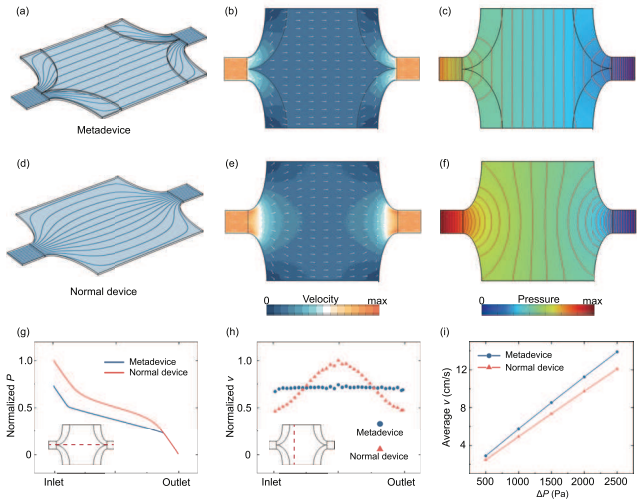

Fig. 2: Steady-state flow field simulation results (on the $z = 0.5$ mm plane). (a)–(c) Simulated streamlines, velocity, and pressure distributions of the metadevice. (d)–(f) Simulated streamlines, velocity, and pressure distributions of the normal device. (g) Normalized pressure distributions at $z = 0.5$ mm, $y = 4$ cm. (h) Normalized velocity distributions at $z = 0.5$ mm, $x = 3.5$ cm. (i) Average flow velocities of the two devices under varying pressure differentials. The data were obtained from the plane at $z = 0.5$ mm, with ethylene glycol (viscosity coefficient of 10 mPa·s) used as the working fluid.

we discuss various fluid types and a range of flow velocities to demonstrate the robustness of our model across a wide spectrum of Reynolds numbers. Additionally, we evaluated the device's robustness to fabrication tolerances in channel height (*e.g.*, ±5% deviations from the nominal $h_2 = 1.59$ mm); simulations indicate that flow uniformity and drag reduction remain effective within these limits, as detailed in sect. S3 of the SM with parametric results.

Conversely, the normal device exhibits typical issues associated with sudden expansion. Flow separation readily occurs at the sharp corners of the expansion section, resulting in large recirculation (vortex) regions near the sidewalls. A central jet persists and only gradually diffuses downstream —requiring a considerable distance for the flow to fill the channel width within the practical device length— leading to a highly non-uniform velocity distribution. The pressure field displays complex patterns associated with these flow features, resulting in substantially greater overall pressure loss.

Comparison of figs 2(c) and (f), under the same colorbar, reveals that the inlet pressure of the two devices differs markedly, with the metadevice exhibiting a significantly lower overall pressure drop. Figure 2(g) presents the extracted pressure profiles at $z = 0.5\,\text{mm}$, $y = 4\,\text{cm}$ for both devices, illustrating the pressure drop across different segments. For clarity, the pressure is normalized as $P/P_{\max}$, where $P_{\max}$ denotes the inlet pressure of the normal device. The results show that the total pressure drop across the metadevice is substantially smaller than that of the normal device. Furthermore, the pressure distribution

[1]Additional information, including theoretical calculations, extended simulation results, and experimental data, can be found in the Supplementary Material and Supplementary Videos.

within the metadevice is linear across all three sections —the inlet channel, central chamber, and outlet channel— exhibiting characteristics typical of pressure drop along a uniform flow path. No additional pressure drop is observed at the transitions, indicating that the local head loss induced by conduit expansion approaches zero. This feature cannot be achieved in conventional expansion-contraction channels.

Figure 2(h) provides the normalized velocity ($v/v_{\max}$, where $v_{\max}$ denotes the maximum velocity of the normal device) profiles at $z = 0.5\,\mathrm{mm}$, $x = 3.5\,\mathrm{cm}$ for both devices, demonstrating that the flow field within the central chamber of the metadevice is highly uniform, whereas the normal device exhibits pronounced non-uniformity in velocity distribution within the chamber. This distinction is critical for certain applications, such as liquid-cooling heat sinks, where velocity non-uniformity can lead to uneven heat dissipation.

For thermal management, not only is a more uniform temperature distribution desirable, but the overall cooling capacity is also crucial; this is closely related to the fluid velocity within the liquid-cooling plate. Generally, with a water pump of the same power, lower flow resistance results in higher fluid velocity and, consequently, improved cooling performance. Here, we apply identical inlet-outlet pressure differences to both the metadevice and the normal device, and compare their average velocities under various pressure gradients, as shown in fig. 2(i). The results indicate that the metadevice achieves a higher average velocity, suggesting greater potential for enhanced heat dissipation.

Experimental validation of the metadevice flow field. – We have designed an experimental setup to empirically validate the flow shaping capabilities of the proposed device. The geometric parameters of the apparatus are identical to those used in the simulations, with the only distinction being the addition of a 10 cm high water reservoir at the inlet, as shown in fig. 3(a). This reservoir serves to visually indicate the inlet pressure via the liquid height, thereby facilitating a straightforward comparison of flow resistance between different devices. The base plate of the entire setups is fabricated from aluminum alloy, while the remaining components are constructed from acrylic panels to facilitate convenient observation of the flow within the channel.

To directly compare the flow resistance between the two devices, we used two pumps to simultaneously inject equal flow rates of liquid into each device and measured the resulting inlet-to-outlet pressure difference. Given the identical outlet boundary conditions, it suffices to compare only the inlet pressures, which are intuitively reflected by the liquid surface heights. Figures 3(b) and (c) show the liquid height comparisons after experiments with dyed water (to facilitate visualization) and artificial blood (4 mPa·s), respectively, with the inlet flow rate maintained at 2 ml/s for both cases. The results indicate that, for both

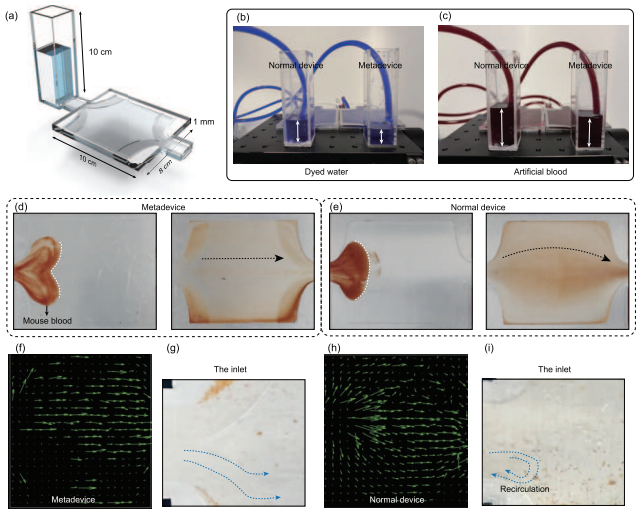

Fig. 3: Experimental setup and results of flow field analysis. (a) Schematic illustration of the experimental setups. (b) and (c): comparison of inlet liquid levels between the two devices under identical flow rates, using dyed water and artificial blood, respectively, as working fluids. (d) The flow behavior of mouse blood upon entering the metadevice, including both the initial and stabilized phases. (e) The flow behavior of mouse blood upon entering a normal device. (f) Velocity field map of particles in the central chamber of the metamadevice, showing straight particle trajectories. The complete experimental results are provided in the Supplementary Video `SV1.mp4`. (g) Enlarged view of the metadevice's inlet, where blue lines indicate the trajectories of the particles. (h) Velocity field map of particles in the central chamber of the conventional device, showing curved particle trajectories. (i) Enlarged view of the normal device's inlet, where flow separation occurs at the corners and particle recirculation is observed.

liquids, the liquid level in the metadevice is significantly lower than that in the conventional device, demonstrating the robust and pronounced flow resistance reduction enabled by the proposed design.

To compare flow field uniformity, mouse blood was injected into the inflow stream, and its distribution was observed in both the metadevice and the normal device. In fig. 3(d), the left panel depicts the initial entry of the blood flow, exhibiting a "fish-tail" pattern that demonstrates effective dispersion by the metadevice. As the blood passes through the metadevice, it rapidly diffuses and forms a smooth, uniform flow profile within the channel, indicating rapid homogenization. In contrast, fig. 3(e) shows that, after injection (left panel), the blood remains concentrated and forms an irregular, non-uniform distribution downstream (right panel). This suggests inferior mixing and insufficient flow uniformity compared to the metadevice. These visualizations clearly indicate that the metadevice enables faster and more homogeneous blood dispersion within the flow field —a crucial factor for blood management, since uniform laminar flow is more conducive to the normal physiological function of endothelial cells.

Furthermore, we introduced microparticles into the fluid and monitored their trajectories to analyze the flow field, a method that may also shed light on the motion of red blood cells in blood. The complete experimental results are provided in the Supplementary Video `SV1.mp4`. As illustrated in figs. 3(f) and (h), we employed the particle image velocimetry tool PIVlab for flow field analysis. The results indicate that within the metamaterial device, particles traverse the expansion chamber along straight paths with uniform velocities. In contrast, within the conventional device, particle trajectories are curved, revealing a non-uniform flow field with higher velocities near the channel centerline and lower velocities near the sides. Additionally, an intriguing phenomenon was observed near the inlet of both devices. In the conventional device, a pronounced recirculation region develops at the inlet —consistent with our simulation predictions— significantly increasing flow resistance, as shown in fig. 3(i). By contrast, no such recirculation occurs at the inlet of the metamaterial device, as depicted in fig. 3(g). If analogous flow phenomena were to arise in blood vessels, they could be extremely detrimental. A detailed analysis of particle motion at the inlet is provided in sect. S4 of the SM.

Experimental validation of the thermal management performance of the metadevice. – Given that the expansion-contraction metadevice proposed in this study can rapidly achieve both uniform flow distribution and low flow resistance, we propose integrating it with the design of liquid cooling plates to enable more efficient and homogeneous heat dissipation. To validate this concept, we fabricated two sets of type-I liquid cooling plates using aluminum alloy: one set incorporates the metadevice at the inlet and outlet, as described in this work, while the other set features a conventional design. The geometric parameters of the flow channels in both plates are kept consistent with those used in the previous fluid dynamics experiments. For the thermal testing, a copper plate with a width of 5 cm serves as the heat source. During the experiment, the copper plate is placed on a heating platform with the liquid cooling plate positioned on top. An infrared camera is used to monitor the temperature distribution across the surface of the cooling plate, as illustrated in fig. 4(a).

Two experiments were conducted under different inlet flow boundary conditions: equal pressure (reflecting constant pump power driving fluid in liquid cooling plates) and equal flow rates.

For equal inlet pressure, transient cooling results (fig. 4(b)) show that both devices were activated simultaneously. The metadevice-integrated cooling plate achieved more uniform heat dissipation from the start, with low-temperature liquid quickly distributing evenly across the cooling area. In contrast, the normal device exhibited faster heat dissipation at the center than at the sides. At equilibrium, the metadevice plate reached lower surface temperatures, attributed to its reduced flow resistance,

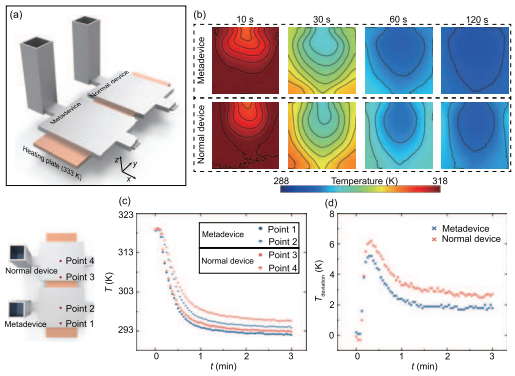

Fig. 4: The liquid cooling plate integrated with the metadevice design exhibited superior heat dissipation performance in validation experiments. (a) Schematic of experimental setup: two liquid cooling plates are placed on a 333 K copper plate for the cooling experiment, with liquid drawn from the same ice-water mixture reservoir. The channel geometry matches the simulation, and the casing is made of aluminum alloy. (b) Transient cooling performance after simultaneous activation of both devices. (c) Temperature-time graphs. Points 1 (T_1) and 2 (T_2) are from the metadevice, points 3 (T_3) and 4 (T_4) are from the normal device group. (d) Temperature difference at center and edge. For the metadevice group, $T_{\text{deviation}} = T_2 - T_1$; for the normal device group, $T_{\text{deviation}} = T_4 - T_3$. A lower $T_{\text{deviation}}$ indicates better temperature uniformity.

which enables faster flow rates and thus lower final temperatures compared to the normal device. The complete cooling process is detailed in the Supplementary Video `SV2.mp4`. Results for equal inlet flow rates are provided in the SM, sect. S5 and Supplementary Video `SV3.mp4`.

We quantitatively compared cooling efficiency by selecting four points at the centers and edges of the two plates and extracting their temperature variations over time (fig. 4(c)). The graph clearly demonstrates that points on the metadevice plate consistently exhibit lower temperatures at identical positions. At the center, temperatures are 1.3 K lower at their lowest, 0.8 K lower when stabilized, and 0.88 K lower on average compared to the normal device. At the edges, temperatures are 2.5 K lower at their lowest, 1.6 K lower when stabilized, and 1.8 K lower on average. Additionally, an important finding is that the metadevice provides superior uniform cooling ability. We compared center and edge temperatures for both devices (fig. 4(d)), where a lower temperature difference indicates better uniformity. The difference between points 1 and 2 (metadevice) was consistently smaller than between points 3 and 4 (normal device), eventually stabilizing at around 1.8 K. Compared to the normal device's difference of approximately 2.6 K, the metadevice reduced the temperature disparity by nearly 30%, indicating improved temperature uniformity.

Conclusion and discussion. – We introduce a metafluidic shaping strategy, based on scattering cancellation theory, to overcome flow distortion and energy

dissipation at abrupt geometric transitions. Our experimentally validated metadevice simultaneously suppresses detrimental recirculation and significantly reduces drag, thereby establishing a remarkably uniform velocity profile and minimizing local head loss.

The device's utility is demonstrated in a liquid cooling system, where its integration achieves superior thermal management, characterized by lower and more homogeneous temperature distributions. This result holds significant promise for enhancing energy efficiency in applications like microchip and electric vehicle cooling. The underlying physical concept is broadly applicable; in microfluidics, for instance, it offers a robust method to ensure uniform reagent delivery and improve process control in platforms for chemical analysis or cell culture [41,42].

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Data availability statement: The data that support the findings of this study are available upon reasonable request from the authors.

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